A LIDAR-Based Approach to Derivative Map Production and Visualizing Bedrock Topography Within The Towns of Proctor and Pittsford, Vermont, USA

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BACKGROUND

This project set out to create isopach and potentiometric surface maps to serve as a value-added derivative map products (Gale et al. 2014). These products were traditionally created using hand-drawn contours and spatially distributed data collected at points. These points included bedrock outcrops and well locations. In this project, we demonstrate how creating a continuous raster data layer can be expedited and automated using geographic information system (GIS) based interpolation and extrapolation techniques.

The use of these interpolation and extrapolation functions with point data presents a powerful opportunity to visualize subsurface geologic relationships. While these techniques have been used to characterize subsurface features for many years, improvements in GIS software offer the ability to use powerful geostatistical tools and robust 3D capabilities which can facilitate the production of three-dimensional visualizations using spatially distributed data. The GIS approach of this project is useful for creating derivative maps that depict surficial overburden, bedrock topography, and potentiometric surfaces in a more expedient method than what would otherwise be used.

This project focused on depicting overburden and potentiometric surface in the southern half of the Proctor quadrangle, containing the area primarily occupied by the towns of Prcotor, Pittsford, and West Rutland. Proctor is located in south-central Vermont within a north-south trending valley primarily underlain by Paleozoic rocks of the Shelburne, Danby, Winooski, Monkton, Dunham, and Dalton formations (Figure 1). There are rare outcroppings of Ordovician Hortonville and Bascom formations. Glacial till dominates the majority of the surficial geology, though there are extensive deposits of lacustrine sediment, kame terraces, and isolated exposures of kame deposits, alluvium, and colluvium (Figure 2).

Elevation within the Proctor quadrangle ranges between approximately 96-643 meters (314-2110 feet) above mean sea level with the greatest relief observed in the central part of our area of interest. The area is characterized by the Taconic and Green Mountain ranges bordering West Rutland from neighboring towns Castleton and Proctor, with relatively lower elevations separating Proctor and Pittsford (Figure 5).

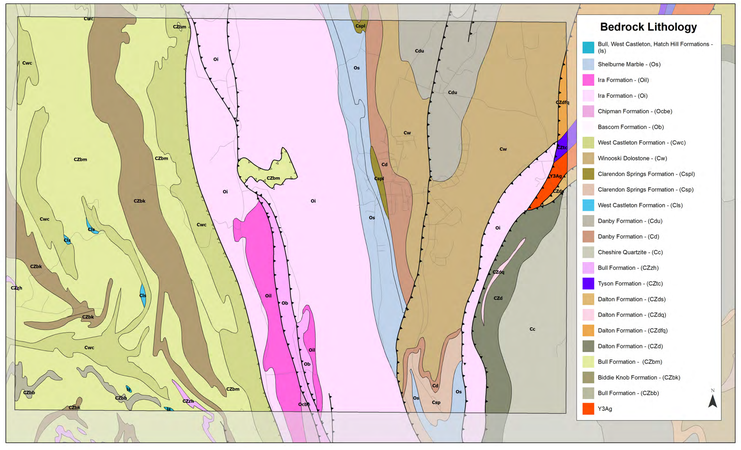


Figure 1: A map (1:100,000 scale) illustrating the spatial distribution of bedrock geology throughout the study area (from Ratcliffe et al. 2011).

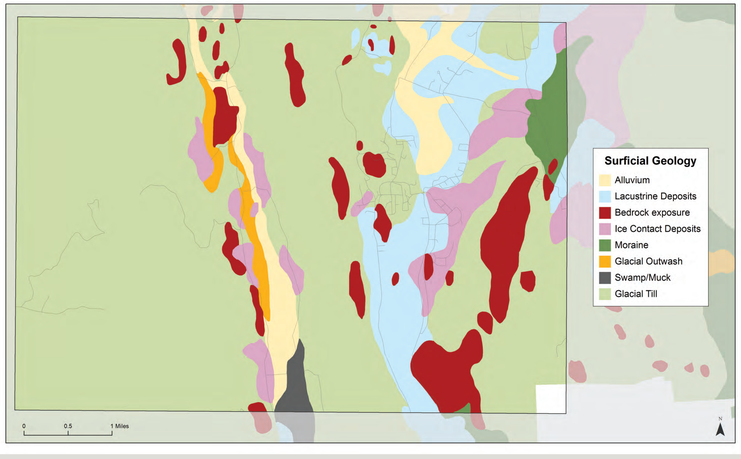


Figure 2: A simplified map (1:250,000) illustrating the spatial distribution of surficial geology throughout the study area (from Doll, 1970).

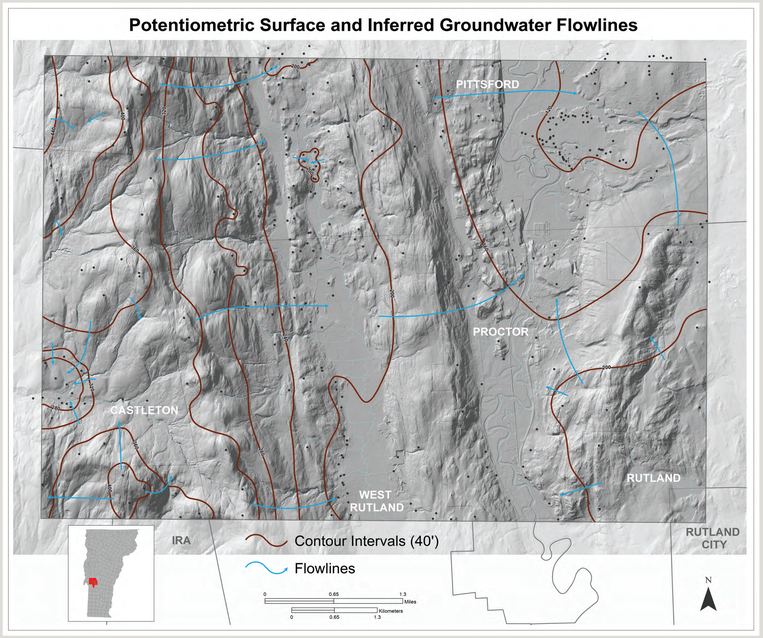


Figure 3: Example of 40' contours derived from an IDW raster surface created using the Geostatistical Analyst.

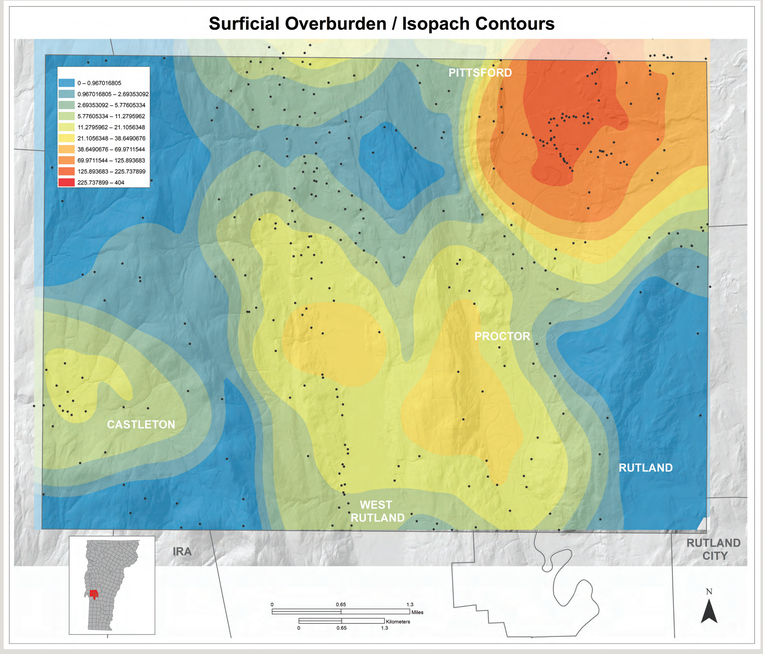


Figure 4: Example of filledcontours derived from a raster surface created using kriging via the Geostatistical Analyst.

METHODOLOGY

To more effectively characterize subsurface hydrologic characteristics within a GIS, 281 private, municipal, and exploratory boring wells were spatially rectified and merged with coordinates for 184 bedrock outcrops were collected using an iPhone 7 using Fulcrum App. Information about well depth, overburden, underlying bedrock formation and depth to water table was extracted from rectified well logs. Some wells were excluded based on a lack of well location information. Bedrock exposures were used with well log data to create continuous raster layers representing surficial overburden and an approximate potentiometric surface. All raster analyses were performed using a 0.7-m digital elevation model (DEM) obtained from the Vermont Center for Geographic Information (VCGI). A raster layer representing the thickness of surficial sediments (also known as an overburden or isopach map) was created using: (1) surficial overburden information from well logs; (2) bedrock exposures identified during field mapping; and (3) bedrock outcrops mapped by previous workers. The ESRI® ArcGIS® Geostatistical Analyst extension was used to evaluate the best kriging function to facilitate the process of estimating overburden and creating a raster layer which covers the entire map area instead of just the areas with wells following Van Hoesen (2014). Ordinary kriging was used for this data. The J-Bessel variogram model was chosen as the best model because it provided the best fit based on prediction error results. The contour list tool in the ArcGIS® Spatial Analyst toolbox was used with the resulting grids to produce contours, which were smoothed by a value ranging from 5-10.

A potentiometric surface was created (Figure 3) using static water levels recorded in well logs and a 0.7-meter DEM. Depth to static water was subtracted from the DEM cell directly beneath the well location and joined to the attribute table to identify the elevation of water within each well. In order to facilitate the process of producing a potentiometric surface and creating a surface covering the entire map area and not just the area with wells, this surface was created using inverse distance weighting (IDW) and the resulting raster layer was used to create 40-foot contours. Groundwater flow lines were manually drawn perpendicular to potentiometric contours to indicate potential flow along a hydraulic gradient within an aquifer. The isopach map (Figure 4) indicates thickest overburden - unsurprisingly - occurs in valley bottoms.

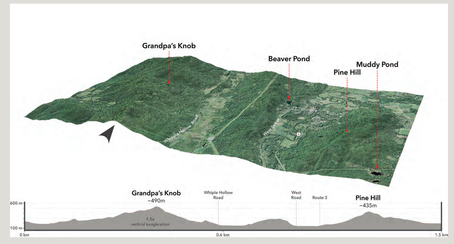


Figure 5: An illustration of the topographical variation throughout the study area visualized by extruding imagery from the National Agriculture Imagery Program (NAIP) with a 0.7-m DEM using ArcScene.

RESULTS

The isopach data is not normally distributed and is strongly weighted towards thinner overburden and bedrock exposures. Bedrock topography generally imitates the landscape, especially in areas of thin overburden. The interpolated isopach map is consistent with both well data and the spatial extent of known surficial units. The areas of thickest overburden are in the northeast corner of the study area. There are a number of areas with limited well data. An abundance of bedrock points was useful in improving the final distribution of overburden. The interpolated potentiometric surface is consistent with well data, surficial geology, and bedrock topography. The hydraulic gradient is gentle and the contours are widely spread apart. Therefore, uncertainty exists in the inferred flow path in some areas of the map. However, there is a general trend of water flowing to the northeast in the areas west of Castleton and water flowing to the west in the Castleton-area and further west. The raster layers produced using well log data are useful, expedited alternatives that help quickly characterize surficial overburden and the hydraulic gradient of an area. The availability of these powerful geostatistical tools within modern GIS software enables the production of derivative maps. Although this project focuses specifically on creating isopach and potentiometric surface products, this methodology could be applied to a variety of different subsurface explorations.

REFERENCES

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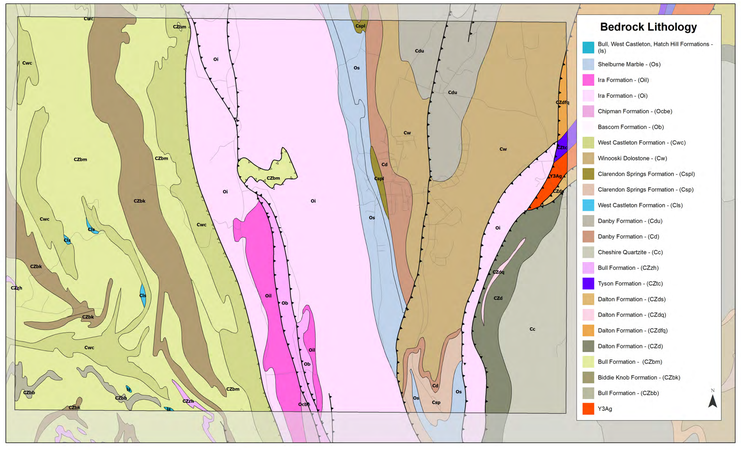


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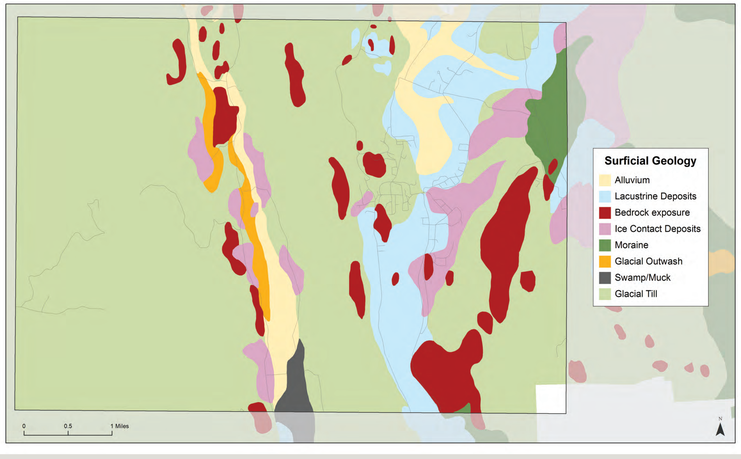


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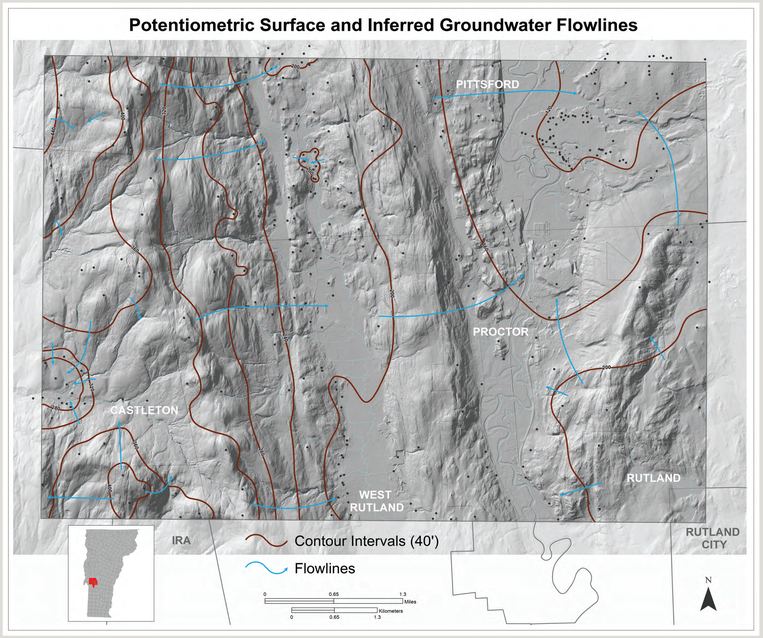


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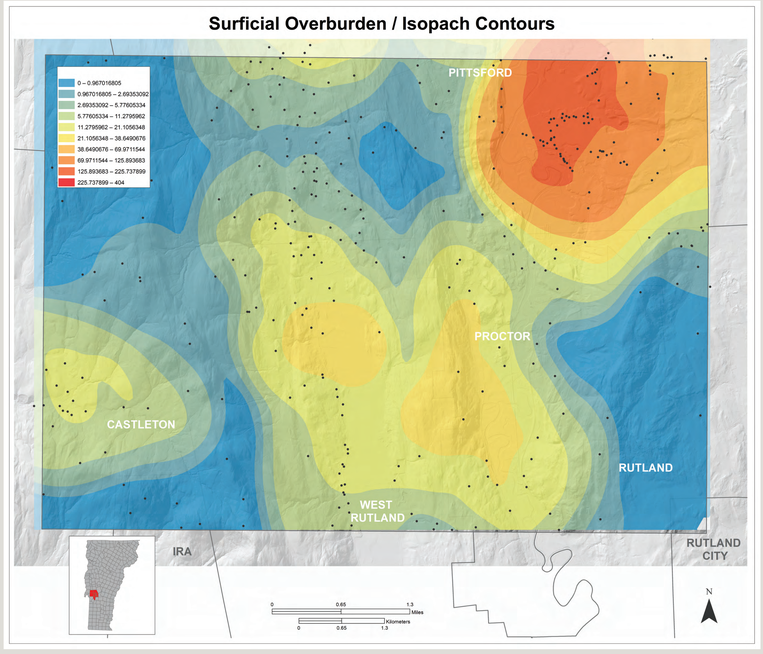


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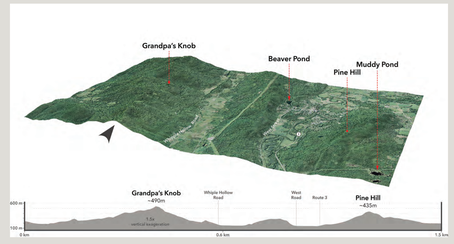


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